Iván Cárdenas-León

Geospatial Specialist · PhD Candidate

linkedin.com/in/ivancardenasleon
github.com/ivan-cardenas/
ivanl.cardenasleon@gmail.com
+31 686185983

19th of October, 2026
Ivar Kolvoort
Head of AI Living Lab
Universiteit van Amsterdam
Amsterdam, Netherlands

Job Application for AI Expert

Dear Ivar Kolvoort,

I am a final-year PhD candidate in Urban Digital Twins at the ITC Faculty of the University of Twente, and I would like to apply for the position of AI Expert at the AI Living Lab. With eight years of experience as an Environmental Engineer and researcher, I combine strong coding skills with a habit of testing new AI tools on real research problems, and I enjoy helping others do the same.

In my current research I am developing an open-source Urban Digital Twin platform across three urban sectors: drinking water supply, heat stress, and housing provision. Alongside this, I have built operational tools: a bibliographic reference enricher through CrossRef, OpenAlex and Semantic Scholar; a Streamlit web application for Circular Energy scenario analysis; and a Python Panel monitoring dashboard for Vitens, the Dutch drinking water authority. I build these with LLM assistants and agentic tools such as Claude Code, Codex and Windsurf, which have drastically reduced the time I spend writing code and given me more time to think about the questions and problems my research is answering. Likewise, I keep myself up-to-date with faster literature search using tools like Asta and I am currently learning the specifics of Elsevier's Leapspace.

To make this repeatable, I have built my own skills and plugins for prototyping, including UX design, software architecture, knowledge-graph building with Graphify, RAG, speech-to-text and deep research assistants. I validate AI-generated code and outputs with test cases and manual review, and I document where AI was used so that the work stays reproducible. This setup not only saves me time, but has also been a great learning experience, and I am excited to keep learning and to share my knowledge with other researchers.

For three years I have lectured on Digital Twins and Urban Heat Modelling at ITC and Vrije Universiteit Amsterdam, which taught me to explain complex methods to students and researchers from different backgrounds. I want to bring that to the AI Living Lab by running hands-on workshops, mentoring students and supporting researchers with prototyping and the ethical use of AI tools in their work. I am particularly drawn to the Lab because how it work across different faculties across UvA and is a great opportunity to learn and support top researchers in their own field while exploring how other disciplines can adapt to and benefit from the power of AI tools in making better questions, improving methodologies and accelerating research. I also see generative tools such as NanoBanana, GPT Image, Claude Design, Google Vids, Higgsfield and Video tools trough ComfyUI as useful for improving science communication and teaching, an area I would like to explore with researchers.

I would welcome the opportunity to discuss how my skills and experience can contribute to the success of the AI Living Lab.

Sincerely,

Iván Cárdenas-León
PhD Candidate in Digital Twins for Urban Planning
ivanl.cardenasleon@gmail.com | linkedin.com/in/ivancardenasleon